CHM 3120: Introduction to Analytical Chemistry

Unit 2 - Section 3:

Statistical Data Treatment and Evaluation

Chapter 5

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Reliability of SD as a Measure of Precision

We've learned that the standard deviation (SD) is a useful indicator of precision; but how reliable is it in practice?

What should we do when we encounter outliers, and how can we identify them?

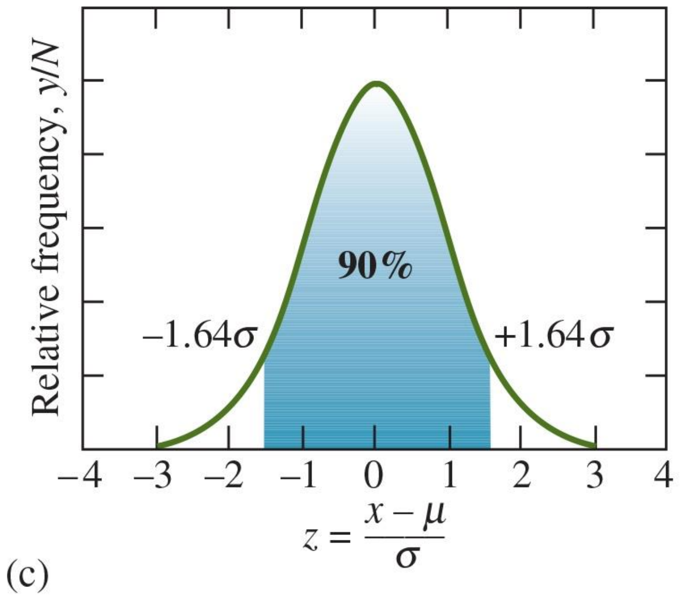
To address these questions, we turn to statistical tests. These tools help us:

- Test hypotheses,
- Construct confidence intervals for our results, and
- Detect and possibly reject outlying data points.

Confidence Intervals

The confidence interval for the mean is a range of values within which the true population mean (μ) is expected to fall, with a certain level of probability.

- The boundaries of this range are called the confidence limits (CL).
  - CL is the probability that the interval actually contains the true mean; typically expressed as a percentage (e.g., 95%).



For a result x with standard deviation σ:

$$\mu = \bar{x} \pm 1.64\sigma$$

This means that if we repeated the experiment many times, then in 90 out of 100 cases, the true mean would lie within the interval range.

Confidence Intervals: T-test

limitations in time or in the amount of available sample prevent us from making enough measurements to assume s is a good estimate of σ. Therefore, comparisons need to be done between different measurements.

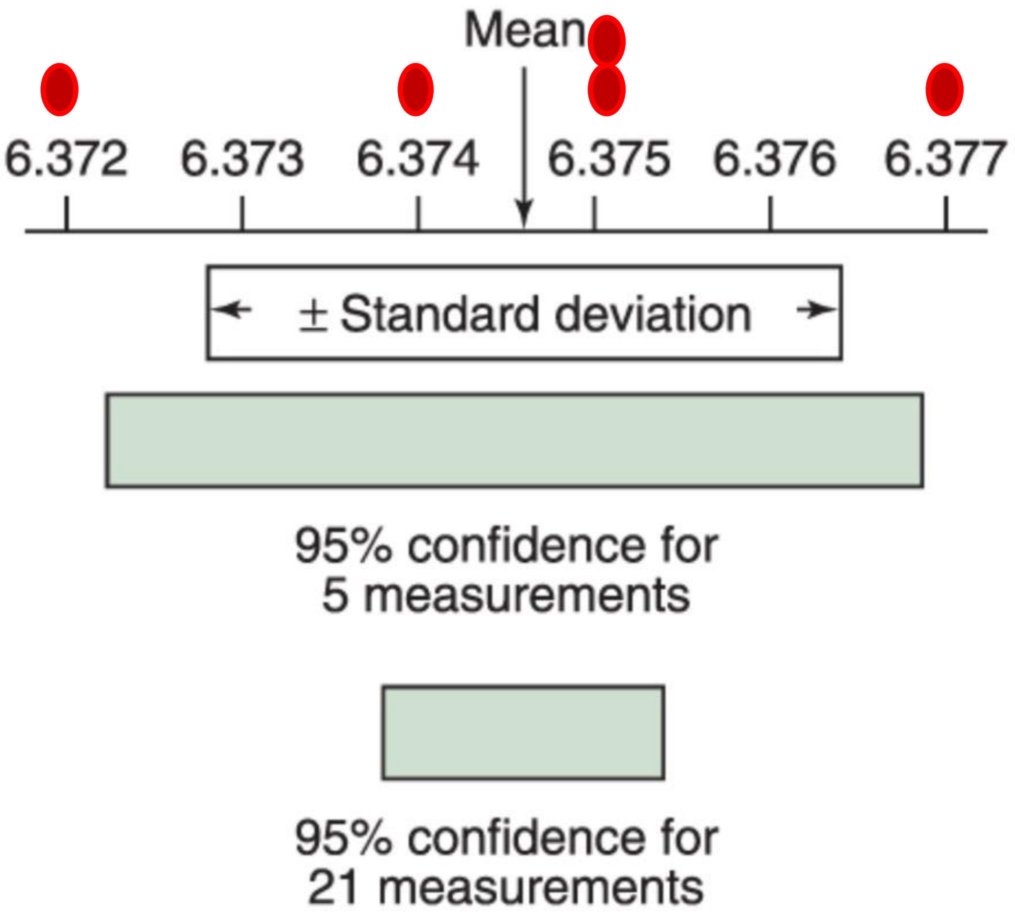
This is done with the Student's T-test and confidence intervals

$$t = \frac{|\bar{x} - \mu|}{s/\sqrt{N}}$$

$$\text{Confidence Interval} = \bar{x} \pm \frac{ts}{\sqrt{N}}$$

Standard Deviations and Confidence Intervals

Data: 6.375, 6.372, 6.374, 6.377 , 6.375



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$$\bar{x} = 6.3746 \pm 0.0018$$

N=5

The Origin of the Student's T-test

VOLUME VI MARCH, 1908 No. 1

BIOMETRIKA.

THE PROBABLE ERROR OF A MEAN.

BY STUDENT.

Introduction.

ANY experiment may be regarded as forming an individual of a "population" of experiments which might be performed under the same conditions. A series of experiments is a sample drawn from this population.

Now any series of experiments is only of value in so far as it enables us to form a judgment as to the statistical constants of the population to which the experiments belong. In a great number of cases the question finally turns on the value of a mean, either directly, or as the mean difference between the two quantities.

William Sealy Gosset
1876 - 1937

$t_{critical}$ vs t_{calc}

df	50%	80%	90%	95%	99%	99.9%
1	1	3.078	6.314	12.71	63.66	636.62
2	0.816	1.886	2.92	4.303	9.925	31.599
3	0.765	1.638	2.353	3.182	5.841	12.924
4	0.741	1.533	2.132	2.776	4.604	8.61
5	0.727	1.476	2.015	2.571	4.032	6.869
6	0.718	1.44	1.943	2.447	3.707	5.959
7	0.711	1.415	1.895	2.365	3.499	5.408
8	0.706	1.397	1.86	2.306	3.355	5.041
9	0.703	1.383	1.833	2.262	3.25	4.781
10	0.7	1.372	1.812	2.228	3.169	4.587
11	0.697	1.363	1.796	2.201	3.106	4.437
12	0.695	1.356	1.782	2.179	3.055	4.318
13	0.694	1.35	1.771	2.16	3.012	4.221
14	0.692	1.345	1.761	2.145	2.977	4.14
15	0.691	1.341	1.753	2.131	2.947	4.073
25	0.684	1.316	1.708	2.06	2.787	3.725
60	0.679	1.296	1.671	2	2.66	3.46
80	0.678	1.292	1.664	1.99	2.639	3.416
100	0.677	1.29	1.66	1.984	2.626	3.39
1000	0.675	1.282	1.646	1.962	2.581	3.3

How it works:

- Compute the t-value (t_{calc})
- Compare the t_{calc} value to the $t_{critical}$ (based on degrees of freedom and confidence level).

If $t_{calc} > t_{critcal}$:

the difference is statistically significant

Practice Problem

You are verifying the label claim that an energy drink contains 100.0 mg of caffeine per can. To test this, you analyze five independent samples and obtain the following results (in mg): 98.5, 100.1, 99.7, 101.0, 99.9

Using these data, determine whether your experimental results are statistically consistent with the manufacturer's claim at the 95% confidence level.

$\sum x_i = 499.2$ $\quad \bar{x} = 99.84$ $\quad$ From t-table for df = 4, 95% confidence: 2.776

$$s = \sqrt{\frac{(98.5 - 99.84)^2 + (100.1 - 99.84)^2 + \cdots + (99.9 - 99.84)^2}{5 - 1}} = 0.8989\, mg = 0.899\, mg$$

$$t = \frac{|99.84 - 100|}{0.899/\sqrt{5}} = 0.3971 = 0.40$$

Since $t_{calc} = 0.40 < 2.776$, the result is **<u>not</u>** significantly different from the claimed value.

Conclusion: The label claim of 100.0 mg caffeine is **statistically supported** at the 95% confidence level.

Practice Problem

The carbohydrate content of a glycoprotein (a protein with sugars attached to it) is found to be 12.6, 11.9, 13.0, 12.7 and 12.5 wt% (g carbohydrate/100 g glycoprotein) in replicate analyses. Find the 50% and 90% confidence intervals for the carbohydrate content.

$\bar{x} = 12.54$

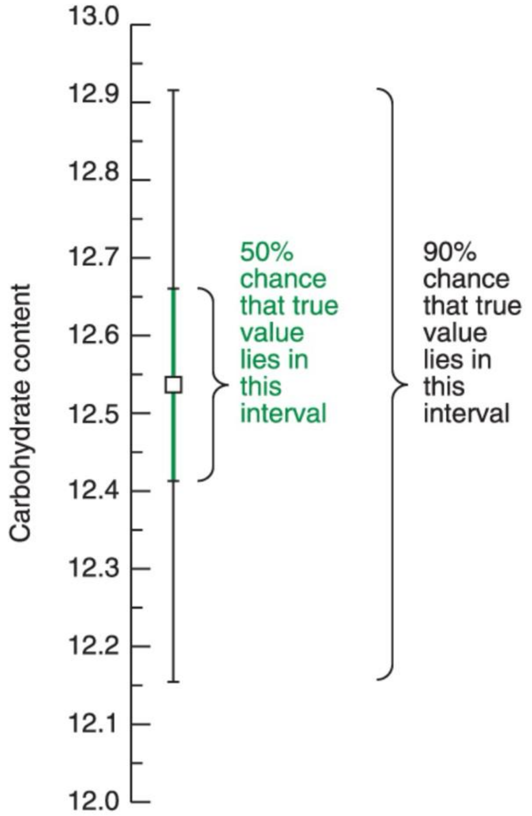
$s = 0.40$ wt%

From t-table for df = 4, 50% confidence: 0.741

From t-table for df = 4, 90% confidence: 2.132

$$50\%\ \text{Confidence Interval} = 12.54 \pm \frac{0.741 \times 0.40}{\sqrt{5}} = 12.54 \pm 0.13\ wt\%$$

$$90\%\ \text{Confidence Interval} = 12.54 \pm \frac{2.132 \times 0.40}{\sqrt{5}} = 12.54 \pm 0.38\ wt\%$$



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Comparison of Means with Student's t

If two sets of measurements are made for the same quantity, the calculated means will usually differ slightly because of random measurement variations.

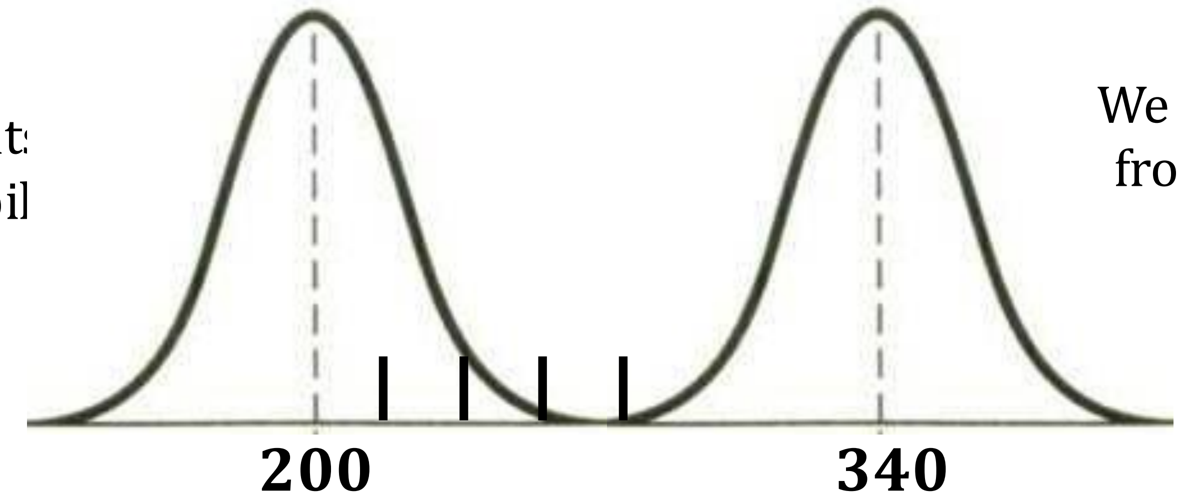
We can **use a t test** to determine whether the observed **difference between the two mean** values is **statistically significant** at a **chosen confidence interval** (95% in most cases).

The test evaluates whether the difference between the two means is larger than expected from random measurement uncertainty (chance) alone.

Statistics therefore provides the probability that the observed difference between two means arises only from random variation.

Comparison of Means with Student's t

Two data distributions:
Suppose two sets of measurement
approximately the same variabil
each: mean= 200 ± 20



We take N = 3 measurements from the second population and calculate a mean.

The question is:
Is the **difference between the two means** large enough that it is unlikely to be caused by random measurement uncertainty?

- If the means are very close, the difference is likely due to random variation.
- As the difference between the means increases, it becomes less likely that the two datasets come from the same population.

The t-test evaluates this.

Comparison of Means with Student's t

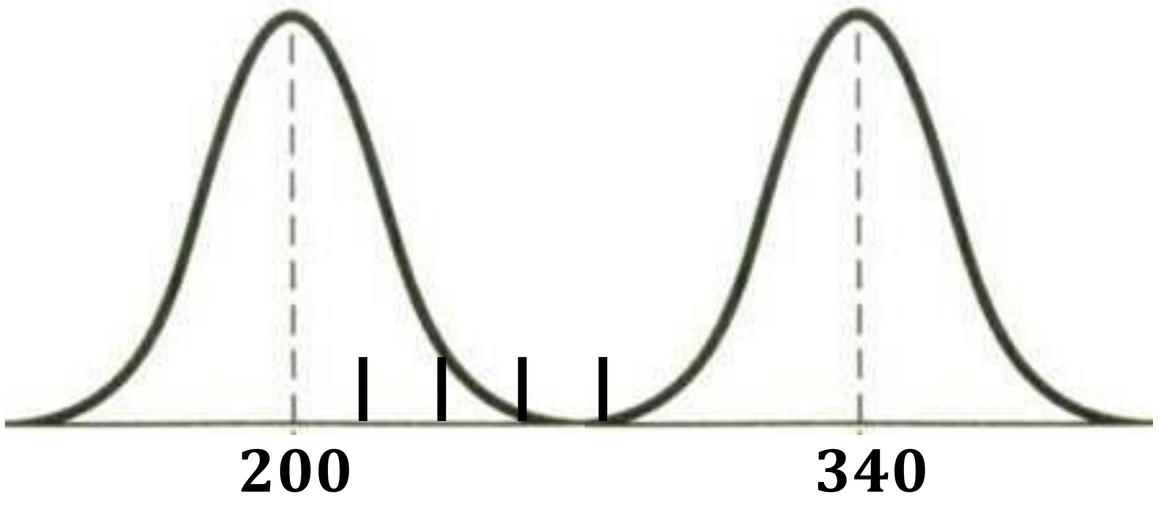
The t statistic quantifies how large the difference is relative to the experimental uncertainty.

- **Large t** → the difference unlikely due to random error
- **Small t** → the difference likely due to random error

Interpretation (for 95% CI):

- If the p-value is **larger** than 0.05 (5%) → the difference **can be explained by random measurement uncertainty**.
- If the p-value is smaller than 0.05 (5%) → the **difference is statistically significant**, and the two datasets likely come from different populations.

In practice, we compare t_{calc} with the critical value from the t table (t_{crit}) for the chosen confidence level and degrees of freedom: if $t_{calc} > t_{crit}$, the difference is statistically significant.



Mean of the second dataset (right peak)	t_{calc}	p-value (difference due to random error)
200	0.000	~1.00
210	0.612	0.58
220	1.225	0.29
230	1.837	0.14
240	2.449	0.07
250	3.062	0.037
340	8.573	~0

Case 1: Comparing a Measurement with a Known Value

3 cases where we handle these comparisons slightly differently:

Case 1: Comparing an Experimental Mean with a Known Value:

Measure a quantity several times, obtaining an average and std dev. We need to **compare our measured value with an accepted (or known) value**.

Does our measured result agree within experimental uncertainty?

$$t = \frac{|\bar{x} - \mu|}{\left(\frac{s}{\sqrt{N}}\right)}$$

This is the case we have discussed so far.

For example, we can use this test to verify a label claim that an energy drink contains 100.0 mg of caffeine per can, or to compare the sulfur content of a coal sample with the certified value from NIST.

Case 1: Comparing a Measurement with a Known Value

Purchase a coal standard from NIST*: contains 3.19 wt% sulfur

Testing a new method:

3.29, 3.22, 3.30, 3.23 $\bar{x} = 3.26$ $s = 0.0408$

Does our answer from the new method agree with the known answer?

- Compute the 95% confidence interval and see if the range includes the known answer
- If the known answer is not within the 95% confidence interval, then the results do not agree

*NIST: National Institute of Standards and Technology

Case 1: Comparing a Measurement with a Known Value

95% Confidence Interval of the Mean

$t_{95\%,3} = 3.182$

$$\text{CI} = \bar{x} \pm \frac{ts}{\sqrt{N}} = 3.26 \pm \frac{3.182 \times (0.0408)}{\sqrt{4}}$$

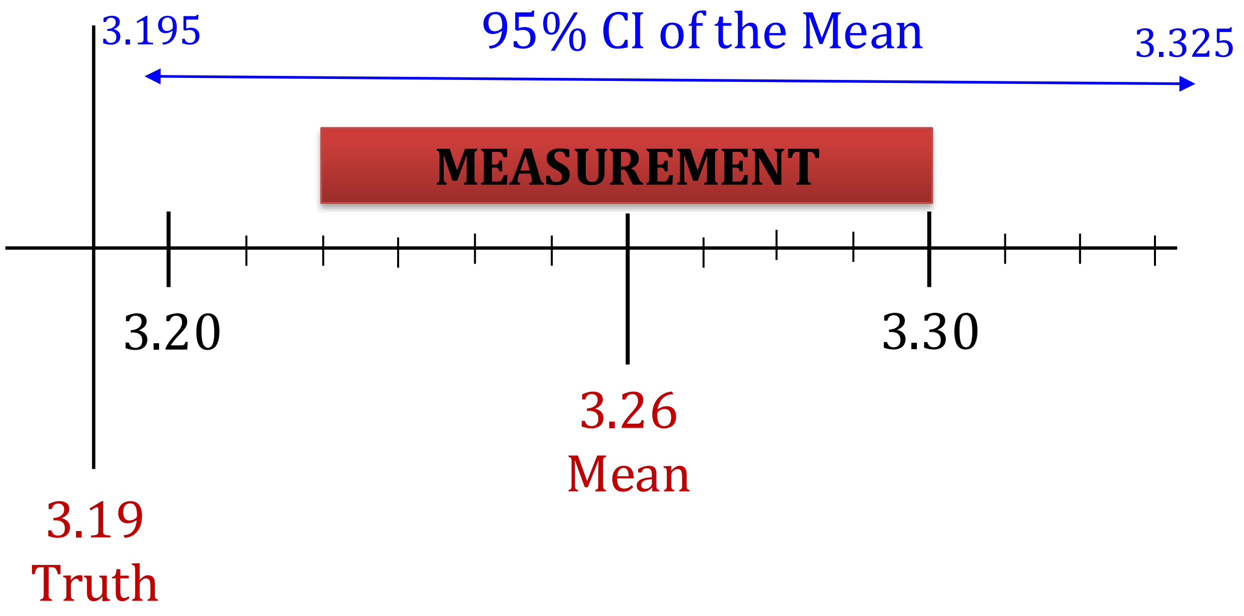
$$= 3.26 \pm 0.065$$

Lower limit = 3.26 – 0.065 = 3.195

Upper limit = 3.26 + 0.065 = 3.325

$$t = \frac{|3.26 - 3.19|}{0.0408/\sqrt{4}} = 3.43$$

Since $t_{calc} > t_{crit}$, the difference is statistically significant, meaning the new method likely has systematic error (bias) relative to the certified value.



Case 2: Comparison of Two Experimental Means

3 cases where we handle these comparisons slightly differently:

Case 2: Comparison of Two Experimental Means:

Measure a quantity several times **by two different methods, getting unique means and std dev** Do the two results agree with each other within experimental uncertainty?

Because both methods **measure the same quantity under similar conditions**, we can assume that their measurement variability is assumed to be similar.

Instead of treating the two SD separately, we combine them into a **pooled SD(s_{pooled})** which combines the two standard deviations to give a better estimate of the common experimental uncertainty when comparing the two means.

Case 2: Comparison of Two Experimental Means

3 cases where we handle these comparisons slightly differently:

Case 2: Comparison of Two Experimental Means:

Measure a quantity several times **by two different methods, getting unique means and std dev** Do the two results agree with each other within experimental uncertainty?

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_{pooled}\sqrt{\frac{N_1 + N_2}{N_1 N_2}}}$$

$$Recall:\ s_{pooled} = \sqrt{\frac{\sum_{i=1}^{N_1}\left(x_{i_1} - \overline{x_1}\right)^2 + \sum_{j=1}^{N_2}\left(x_{j_2} - \overline{x_2}\right)^2 + \sum_{k=1}^{N_3}\left(x_{k_3} - \overline{x_3}\right)^2 + \cdots}{(N_1 - 1) + (N_2 - 1) + (N_3 - 1) + \cdots}} = \sqrt{\frac{s_1^2\,(N_1 - 1) + s_2^2\,(N_2 - 1)}{N_1 + N_2 - 2}}$$

Practice Problem

Measuring bicarbonate (HCO_3^-) in the blood of racehorses (banned substance). We're certifying a new instrument: does the new instrument give the same results as the old instrument?

Is the standard deviation from the new instrument substantially different from that of the original instrument?

	Original Instrument	**New Instrument**
Mean	36.14	36.20
Std dev	0.28	0.47
# measurements	10	4

Practice Problem

$$s_{pooled} = \sqrt{\frac{0.0784\,(10-1) + 0.2209(4-1)}{10+4-2}} = \sqrt{\frac{1.3683}{12}} = 0.338$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_{pooled}\sqrt{\frac{N_1+N_2}{N_1 N_2}}} = \frac{|36.14 - 36.20\,|}{0.338 \times \sqrt{\frac{10+4}{10 \times 4}}} = \frac{0.06}{0.1998} = 0.300$$

$df\colon N_1 + N_2 - 2 = 10 + 4 - 2 = 12 \;\rightarrow$ From t-table for df = 12, 95% confidence: 2.179

$$t_{calc} = 0.300 < t_{95\%,12} = 2.179$$

There is no significant difference in the mean values between the two instruments at the 95% confidence level

Homework

In a forensic investigation, a glass containing red wine and an open bottle were analyzed for their alcohol content in order to determine whether the wine in the glass came from the bottle. On the basis of six analyses, the average content of the wine from the glass was established to be 12.61% ethanol. Four analyses of the wine from the bottle gave a mean of 12.53% alcohol. The 10 analyses yielded a pooled standard deviation s_{pooled}=0.070%. Do the data indicate a difference between the wines?

Answer check: No significant difference at the 95% confidence level

Case 3: Comparing Individual Differences

3 cases where we handle these comparisons slightly differently:

Case 3: Comparing Individual Differences

A sample is measured by two different methods. A different sample is then measured by the same two methods. This is repeated for n different samples, **producing pairs of measurements for each sample**. *Do the two methods agree within experimental uncertainty?*

Because both methods are applied to the same sample, the results are paired.

Instead of comparing the two means directly, we **analyze the difference between the two measurements** for each sample.

Case 3: Comparing Individual Differences

Use two methods to make single measurements on several different samples (no duplicate measurements). Do the 2 methods give the same answer within experimental uncertainty?

$$t = \frac{|\bar{d}|}{\left(\frac{s_d}{\sqrt{n}}\right)}$$

$$s_d = \sqrt{\frac{\Sigma\left(d_i - \bar{d}\right)^2}{n-1}}$$

d_i = Result $_{(\text{Method 2})}$ – Result $_{(\text{Method 1})}$

$$\bar{d} = \frac{\sum d_i}{n}$$

- Be careful when calculating the differences:

  d_i = Result $_{(\text{Method 2})}$ – Result $_{(\text{Method 1})}$
- Positive or negative values simply indicate which method gives the larger result.
- The absolute value |đ| is used in the t-equation.

Practice Problem

A laboratory is evaluating a new, more economical electrochemical method for determining aluminum (Al) concentration in water samples to replace the existing, more costly Atomic Absorption Spectroscopy method. To compare the methods, 11 water samples were analyzed using each method.

Do the two methods give statistically equivalent results at the 95% confidence level?

Sample	Method 1 (μg/L)	Method 2 (μg/L)
1	17.2	14.2
2	23.1	27.9
3	28.5	21.2
4	15.3	15.9
5	23.1	32.1
6	32.5	22.0
7	39.5	37.0
8	38.7	41.5
9	52.5	42.6
10	42.6	42.8
11	52.7	41.1

Practice Problem

Sample	Method 1	Method 2	d_i = Method 2 – Method 1	$(d_i - \bar{d})$	$(d_i - \bar{d})^2$
1	17.2	14.2	-3	-0.51	0.26
2	23.1	27.9	4.8	7.29	53.16
3	28.5	21.2	-7.3	-4.81	23.13
4	15.3	15.9	0.6	3.09	9.55
5	23.1	32.1	9	11.49	132.04
6	32.5	22	-10.5	-8.01	64.15
7	39.5	37	-2.5	-0.01	0.00
8	38.7	41.5	2.8	5.29	27.99
9	52.5	42.6	-9.9	-7.41	54.89
10	42.6	42.8	0.2	2.69	7.24
11	52.7	41.1	-11.6	-9.11	82.98
			$\bar{d}$ = -2.49		$\sum(d_i - \bar{d})^2$ = 455.388

$$s_d = \sqrt{\frac{\Sigma(d_i - \bar{d})^2}{n-1}} = \sqrt{\frac{455.388}{11-1}} = 6.748$$

$$t = \frac{|\bar{d}|}{\left(\frac{s_d}{\sqrt{n}}\right)} = \frac{|-2.49|}{\left(\frac{6.748}{\sqrt{11}}\right)} = 1.224$$

$$t_{calc} = 1.224 < t_{95\%,10} = 2.228$$

There is no significant difference in the mean values between the two methods at the 95% confidence level

Homework

A new automated procedure for determining glucose in serum (Method A) is compared to the established method (Method B). Both methods are performed on serum from the same six patients in order to eliminate patient-to-patient variability. Do the following results confirm a difference in the two methods at the 95% confidence level?

Patient	Method A glucose (mg/L)	Method B glucose (mg/L)
1	1044	1028
2	720	711
3	845	820
4	800	795
5	957	935
6	650	639

Answer check: The two methods are significantly different at the 95% confidence level.

Comparison of Variances: F-Test

The F-test is used to determine whether the variances (or standard deviations) of two data sets are significantly different.

The **variance (S^2)** reflects the precision of a measurement.

Therefore, the F-test is commonly used **to compare the precision** of two sets of replicate measurements or two analytical methods.

Example uses:

- Compare **the precision of two analytical methods** (e.g., UV-Vis vs ICP-OES for measuring Fe^{3+} concentration)
- Evaluate **measurement variability of an instrument** (e.g., repeated GC measurements of caffeine in a sample)
- Determine whether **two analysts obtain similar precision** (e.g., two students performing replicate titrations of acetic acid in vinegar)
- Compare **precision before and after instrument maintenance** (e.g., HPLC retention time variability before and after column replacement)

F-Test

The F test statistic is the ratio of the two sample variances:

$$F_{calc} = \frac{s^2_{larger}}{s^2_{smaller}}$$

The larger variance is placed in the numerator so that F≥1.

If $F_{calc} > F_{critical}$:
the variances are significantly different at the 95% confidence level.

Critical Values of F at the 5% Significance Level (95% Confidence Level)

df for **num**→ ***denom***↓	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**15**	**20**	**∞**
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	245.9	248.0	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.43	19.45	19.50
3	10.13	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.786	8.703	8.660	8.526
4	7.709	6.994	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.858	5.803	5.628
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.722	4.753	4.619	4.558	4.365
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	3.938	3.874	3.669
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.511	3.445	3.230
8	5.318	4.459	4.066	3.838	3.687	3.581	3.500	3.438	3.388	3.347	3.218	3.150	2.928
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.006	2.936	2.707
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.845	2.774	2.538
11	4.844	3.982	3.587	3.257	3.204	3.095	3.012	2.948	2.896	2.854	2.719	2.646	2.404
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.617	2.544	2.296
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.533	2.459	2.206
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.463	2.388	2.131
15	4.534	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.403	2.328	2.066
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.352	2.276	2.010
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.308	2.230	1.960
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.269	2.191	1.917
19	4.381	3.552	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.234	2.155	1.878
20	4,351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.203	2.124	1.843
∞	3.842	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.666	1.570	1.000

Practice Problem

Back to our bicarbonate example…

Instrument	SD	Variance (s^2)	n
Original	0.28	0.0784	10
New	0.47	0.2209	4

$$F_{calc} = \frac{0.2209}{0.0784} = 2.817$$

$$F_{calc} = 2.817 < F_{critical} = 3.863$$

There is no significant difference in variance between the two instruments at the 95% confidence level

Critical Values of F at the 5% Significance Level (95% Confidence Level)

df for **num**→ ***denom***↓	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**15**	**20**	**∞**
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	245.9	248.0	254.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.43	19.45	19.50
3	10.13	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.786	8.703	8.660	8.526
4	7.709	6.994	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.858	5.803	5.628
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.722	4.753	4.619	4.558	4.365
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	3.938	3.874	3.669
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.511	3.445	3.230
8	5.318	4.459	4.066	3.838	3.687	3.581	3.500	3.438	3.388	3.347	3.218	3.150	2.928
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.006	2.936	2.707
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.845	2.774	2.538
11	4.844	3.982	3.587	3.257	3.204	3.095	3.012	2.948	2.896	2.854	2.719	2.646	2.404
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.617	2.544	2.296
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.533	2.459	2.206
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.463	2.388	2.131
15	4.534	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.403	2.328	2.066
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.352	2.276	2.010
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.308	2.230	1.960
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.269	2.191	1.917
19	4.381	3.552	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.234	2.155	1.878
20	4,351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.203	2.124	1.843
∞	3.842	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.666	1.570	1.000

Homework

A standard method for the determination of the carbon monoxide (CO) level in gaseous mixtures is known from many hundreds of measurements to have a standard deviation of 0.21 ppm CO. A modification of the method yields a value for s of 0.15 ppm CO for a pooled data set with 12 degrees of freedom. A second modification, also based on 12 degrees of freedom, has a standard deviation of 0.12 ppm CO.

Is either modification significantly more precise than the original?

Answer check: Modification 1: No significant improvement
Modification 2: Significantly better precision

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A researcher compares a new glucose biosensor with a hospital reference method using samples from the same 4 patients.

Even if both methods have similar variances, why is a paired t-test conceptually better than an unpaired t-test?

Patient	Reference	Biosensor
1	80	85
2	120	125
3	200	205
4	300	305

A. An unpaired t-test requires at least 30 patients.

B. The unpaired test compares overall group means, so the large patient-to-patient variability can hide the small systematic +5 mg/dL bias.

C. Paired t-tests are only needed when two different instrumental techniques are compared.

D. Unpaired t-tests assume there is no random error.

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Patient	Reference	Biosensor
1	80	85
2	120	125
3	200	205
4	300	305

What the unpaired test "sees"

- Reference: 80, 120, 200, 300
- Biosensor: 85, 125, 205, 305

Because glucose levels vary a lot from patient to patient, both data sets look very spread out.

Compared with that large spread, the consistent +5 mg/dL difference between the two methods looks small, so an unpaired t-test may miss it.

The small +5 bias becomes hidden inside that spread.

What the paired test "sees"

The paired test analyzes the differences:

$$d = \text{Biosensor} - \text{Referenced}$$

Patient	Difference
1	+5
2	+5
3	+5
4	+5

The patient-to-patient variability cancels out, making the systematic bias obvious.

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Two analytical methods are compared.

- Method A (very precise): 10.1, 9.9, 10.0, 10.1, 9.9; s_A= 0.1
- Method B (very noisy): 8.5, 11.5, 9.0, 12.0, 9.0; s_B= 1.7

Why is it mathematically dangerous to skip the F-test and directly use a pooled t-test?

A. The t-test cannot be calculated without first finding F_{calc}

B. The F-test measures accuracy while the t-test measures precision.

C. The F-test determines the degrees of freedom for paired t-tests.

D. Pooling assumes both methods have similar variances. If one method is much noisier, the uncertainty estimate becomes distorted.

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The F-test checks whether the two methods have statistically similar variances. Using the given standard deviations:

$$F_{calc} = \frac{(1.70)^2}{(0.10)^2} = \frac{2.89}{0.01} = 289$$ → This is an enormous difference in variance.

That means Method B is much noisier than Method A.

A pooled t-test combines both variances into one shared uncertainty estimate: $s_{\text{pooled}} = \sqrt{\frac{s_1^2\,(N_1-1)+s_2^2\,(N_2-1)}{N_1+N_2-2}}$

But pooling assumes: $s_1 \approx s_2$ which is clearly not true here.

As a result, the noisy method artificially inflates the uncertainty estimate and can lead to misleading statistical conclusions.

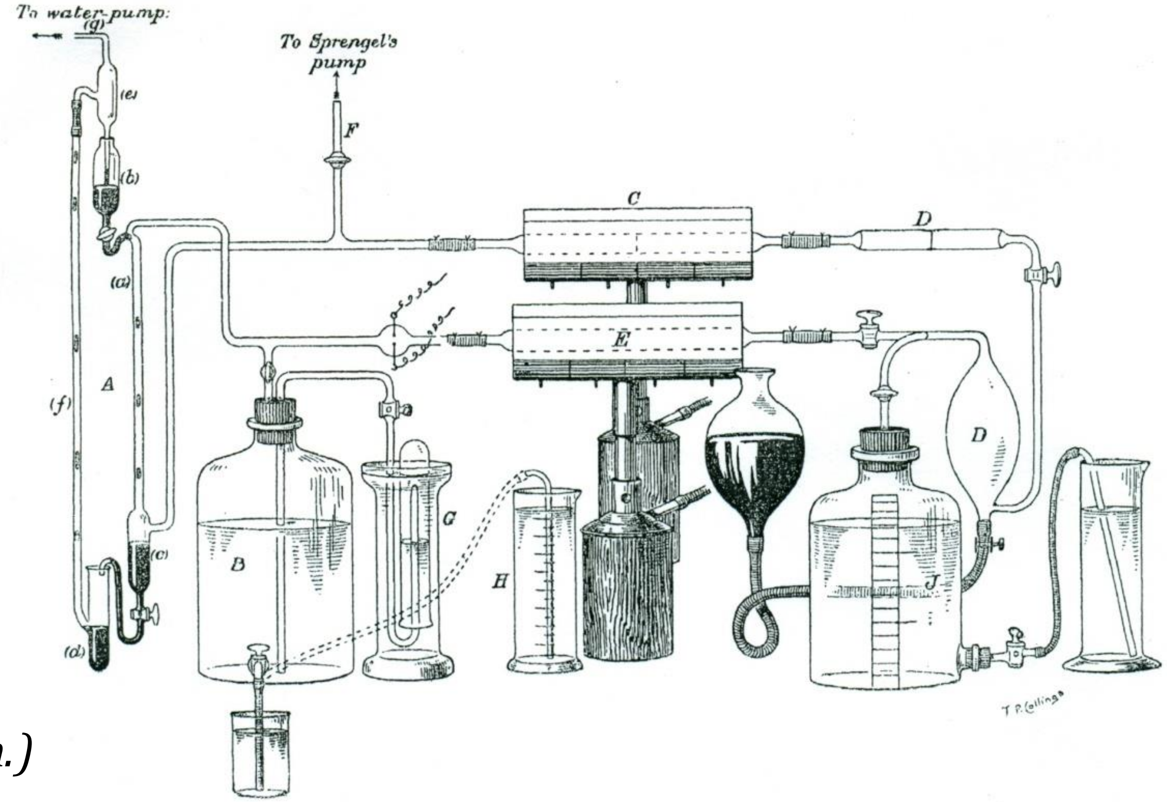
Rayleigh’s Nitrogen Measurements

In the late 1800s, Lord Rayleigh carefully measured the density of nitrogen gas prepared by two different methods.

- Nitrogen obtained from air
- Nitrogen produced chemically

I. “On an Anomaly encountered in Determinations of the Density of Nitrogen Gas.” By LORD RAYLEIGH, Sec. R.S. Received February 16, 1894.

In a former communication* I have described how nitrogen, prepared by Lupton’s method, proved to be lighter by about 1/1000 part than that derived from air in the usual manner. In both cases a red



(This question led to the discovery of a new element: argon.)

Rayleigh's Nitrogen Measurements

Rayleigh noticed a small but consistent difference in the measured masses.

Question:

Is the nitrogen obtained from air actually different from nitrogen produced chemically, or is the difference due to random experimental error?

Measurement #	Nitrogen from Air (g)	Nitrogen from Chemical Reaction (g)
1	2.31017	2.30143
2	2.30986	2.29890
3	2.31010	2.29816
4	2.31001	2.30182
5	2.31024	2.29869
6	2.31010	2.29940
7	2.31028	2.29849
8	—	2.29889
Average	**2.31011**	**2.29947**
Standard deviation	**0.000143**	**0.00138**

Nitrogen from N_2O by Hot Iron.*

December 26, 1893...... 2·29869 }
December 28, 1893...... 2·29940 } Mean, 2·29904

Nitrogen from Ammonium Nitrite passed over Hot Iron.

January 9, 1894 .... 2·29849 }
January 13, 1894 .... 2·29889 } Mean, 2·29869

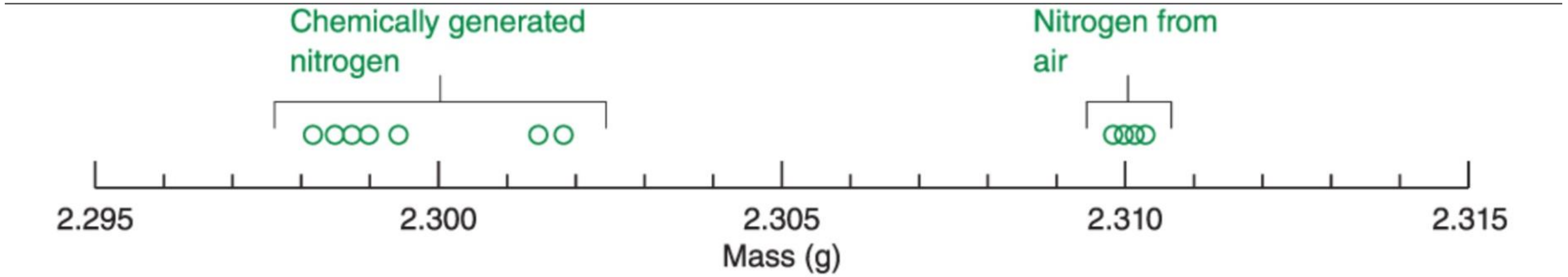
With these are to be compared the weights of nitrogen derived from the atmosphere.

Nitrogen from Air by Hot Iron.

December 12, 1893 ...... 2·31017
December 14, 1893 ...... 2·30986 (H)
December 19, 1893 ...... 2·31010 (H)
December 22, 1893 ...... 2·31001
Mean, 2·31003

Nitrogen from Air by Ferrous Hydrate.

January 27, 1894........ 2·31024
January 30, 1894........ 2·31010
February 1, 1894 .... 2·31028
Mean, 2·31020



This question led to the discovery of a new element: argon.

Interpreting Rayleigh's Results

Two statistical questions arise from these measurements.

1. Are the standard deviations significantly different?
→ **Compare the precision using an F-test**

2. Are the average masses significantly different?
→ **Compare the means using a t-test**

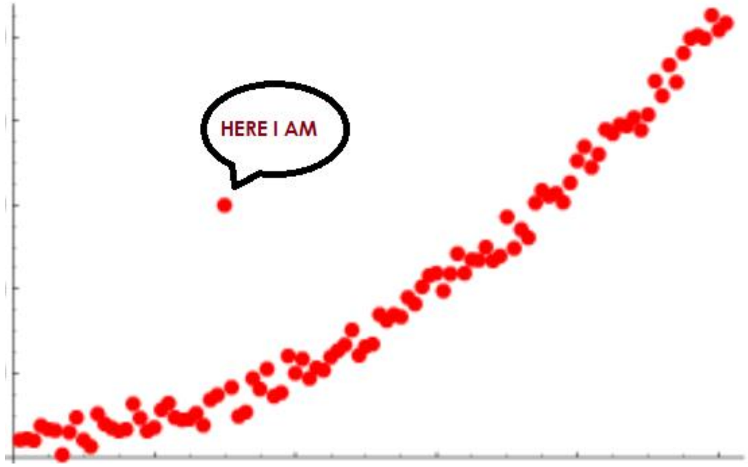
Rayleigh observed that nitrogen obtained from air was consistently slightly heavier than nitrogen produced chemically.

Measurement #	Nitrogen from Air (g)	Nitrogen from Chemical Reaction (g)
1	2.31017	2.30143
2	2.30986	2.29890
3	2.31010	2.29816
4	2.31001	2.30182
5	2.31024	2.29869
6	2.31010	2.29940
7	2.31028	2.29849
8	—	2.29889
Average	**2.31011**	**2.29947**
Standard deviation	**0.000143**	**0.00138**

This discrepancy eventually led to the discovery of argon in 1894, and contributed to Rayleigh receiving the Nobel Prize in Physics in 1904 (Link).

What About the Outlier?

- toWhat is bad data?
- What exactly is an outlier?
- Is it acceptable remove a data point?



An outlier is a data point that falls far from the other measurements.

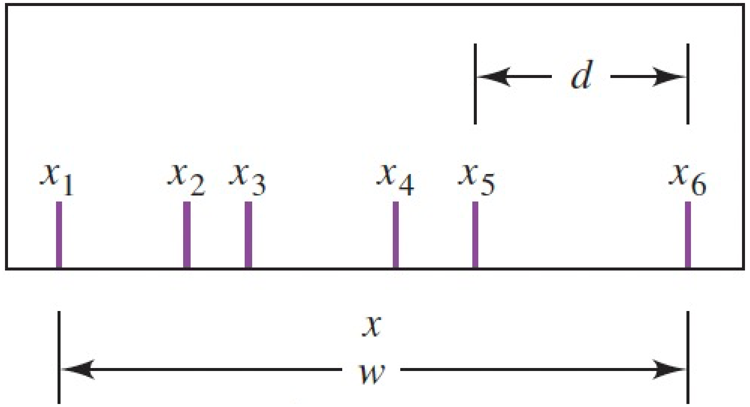
The Q test and Grubbs test help determine whether a suspected outlier can be rejected.

Outliers should only be removed if there is statistical or experimental justification.

Q-Test for Detecting an Outlier

$$Q = \frac{d}{w} = \frac{|x_{suspected} - x_{nearest\ value}|}{w}$$

w: range of the data set (max − min)



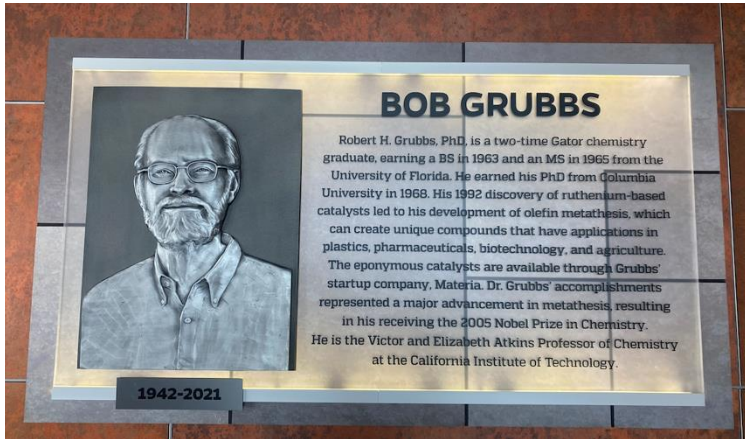
Critical Values of the Q-Test for Detecting Outliers

Number of observations	90% Confidence	95% Confidence	99% Confidence
3	0.941	0.970	0.994
4	0.765	0.829	0.926
5	0.642	0.710	0.821
6	0.560	0.625	0.740
7	0.507	0.568	0.680
8	0.468	0.526	0.634
9	0.437	0.493	0.598
10	0.412	0.466	0.568

If $Q > Q_{crit}$, reject the suspected outlier.

Grubbs Test for Detecting an Outlier

Photos from the UF Chemistry Department honoring **Robert H. Grubbs**, a UF alumnus and Nobel Laureate.



$$G_{calc} = \frac{|x_{suspected} - \bar{x}|}{s}$$

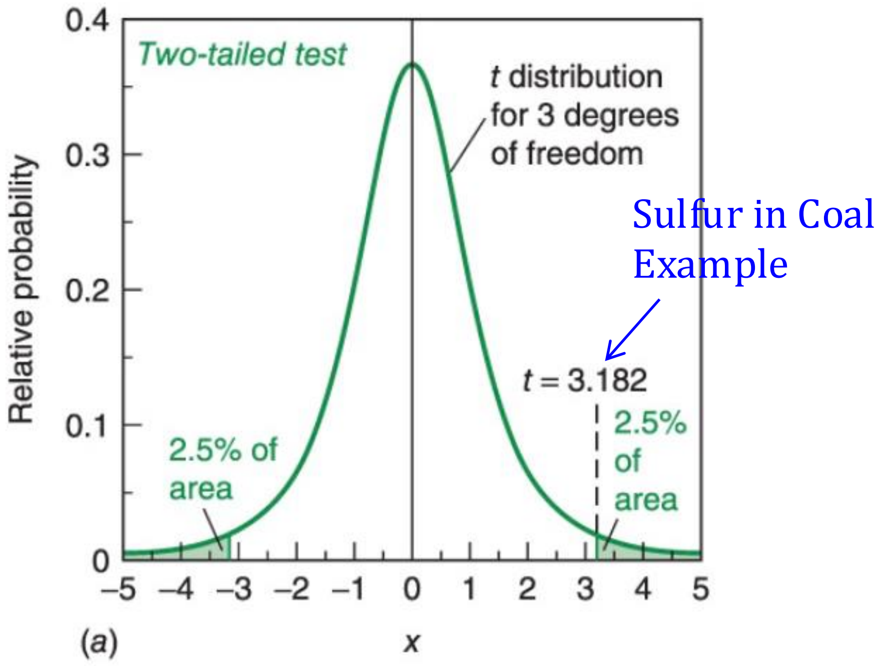
If $G_{calc} > G_{crit}$, the suspected value can be rejected, and the mean and standard deviation **should be recalculated**.

Calculate **s using the entire data set first**, including the suspected value.
Only remove the value after the test confirms it is an outlier.

Critical Values of the G-Test for Detecting Outliers

Number of observations	G (95% confidence)
3	1.153
4	1.463
5	1.672
6	1.822
7	1.938
8	2.032
9	2.110
10	2.176
11	2.234
12	2.285
15	2.409
20	2.557
30	2.745
50	2.956

One-Tailed vs. Two-Tailed Significance Tests



Could the data be significantly different in either direction from the expected value?
Example: Did this coal sample contain more sulfur than the limit, or less sulfur?

So, we split the "rejection area" (our 5% cutoff) into two tails:

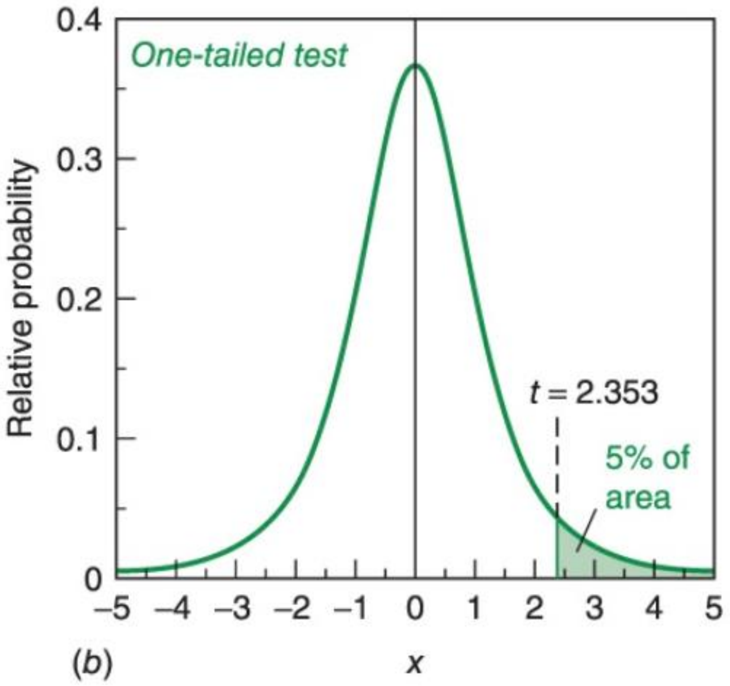
- 2.5% on the left (too low),
- 2.5% on the right (too high)

The cutoff for significance ends up at t = 3.182 (for 3 degrees of freedom in this example).

This is stricter because the 5% rejection region is split between both tails.

Note: In most analytical chemistry applications, two-tailed tests are typically used because deviations can occur in either direction.

One-Tailed vs. Two-Tailed Significance Tests



Here we ask only whether the value is greater than (or less than) the limit.

Example: Does the coal contain more sulfur than allowed by regulation? Now the entire 5% rejection region is in one tail.

That makes it "easier" to declare significance → cutoff is t = 2.353, smaller than in the two-tailed case.

In a one-tailed test, all 5% goes into just one tail.
So instead of using a 95% two-tailed value, we use the 90% one-tailed value.

Use the 90% t value for a one-tailed test.

Note: One-tailed tests are used only when the direction of the difference is specified in advance.

$$t_{calculated} = \frac{\lfloor \bar{x} - regulatory\ limit \rfloor}{s}\sqrt{n}$$